

ALAN BIJU PALAYIL

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Professional Summary

Data Engineer and application developer specializing in the automation, modernization, and cloud migration of high-volume financial data systems. Deep expertise in Python, SQL, and cloud architectures (Azure and AWS) applied to Commercial Mortgage Loan (CML) and Derivatives investment-accounting platforms. Designed and delivered a production JSON message-generation platform that migrated CML accounting off a third-party vendor and now runs daily against 1,500+ active loans. Cybersecurity-trained, Zero Trust-certified, and an actively publishing researcher (ten peer-distributed monographs, 2026) pursuing a Ph.D. in Information Technology focused on secure, data-driven, and explainable AI systems.

Technical Skills

Languages & Frameworks:	Python, SQL, Java, C#, C++, JavaScript, React
Databases & ETL:	PostgreSQL, MySQL, Oracle, Azure Synapse Analytics, Azure Data Lake Storage (ADLS), Markit EDM, Informatica, AppWorx, ETL/ELT pipelines, batch processing
Cloud & DevOps:	AWS (Solutions Architect, SysOps, Developer), Microsoft Azure, Azure DevOps, Git/GitHub, JIRA, Linux, CI/CD
Integration & Formats:	JSON, XML, REST APIs, secure file transfer, enterprise scheduling
Security & Governance:	Zero Trust Architecture (CCZT), Network Security, NIST CSF, Model Risk Management, data validation
AI & Data Science:	Machine Learning, anomaly detection, time-series forecasting, data mining, predictive analytics
Automation & Hardware:	PowerShell, Bash, Arduino/ESP32
Methodologies:	Agile/Scrum, SDLC, functional & technical test-plan development, root-cause analysis

Professional Experience

Data Engineer | Genworth Financial – Richmond, VA **July 2026 – Present**

- Lead data engineering and application modernization for enterprise investment-accounting systems supporting Commercial Mortgage Loan (CML) and Derivatives operations.
- Design, develop, and maintain Python- and SQL-based data pipelines, ETL workflows, interfaces, extracts, reconciliation processes, and JSON-based message-generation services.
- Build cloud-native processing solutions on Microsoft Azure using Synapse Analytics, Azure Data Lake Storage (ADLS), and notebooks.
- Serve as IT owner for CML processing within PAM — technical oversight, enhancement planning, production support, data validation, and operational stability.
- Engineer automated file-transfer, monitoring, retry, notification, dependency-management, and reconciliation solutions that reduce manual intervention and improve reliability.
- Partner with Investments Operations, Accounting, Controllershship, Infrastructure, QA, vendors, and engineering teams; follow Agile, SDLC, version-control, and change-management standards.

Application Development Analyst | Genworth Financial – Richmond, VA **December 2025 – July 2026**

- Led application engineering and modernization initiatives for investment-accounting systems, owning CML workflows within PAM and supporting migration of legacy Derivatives processing to Microsoft Azure.
- Contributed to rehosting and modernization of the PAM Legacy Message Generator and related Derivatives processing components.
- Developed Python applications, SQL queries, ETL workflows, automated interfaces, extracts, reconciliations, and JSON message-generation services.

- Supported CML migration workflows spanning loan, collateral, fee, payment, interface, and reconciliation processing.
- Investigated production incidents, performed root-cause analysis, and implemented sustainable fixes improving reliability, auditability, and user experience.

Associate Application Development Analyst | Genworth Financial – Richmond, VA **July 2023 – December 2025**

- Developed and supported automation, data interfaces, and reporting for enterprise investment-accounting workflows across Fixed Income, CML, and Enterprise Data Management (EDM).
- Authored Python scripts and SQL queries for data transformation, validation, reconciliation, reporting, and recurring operational tasks.
- Built automated reports, extracts, interfaces, ETL workflows, and batch-processing solutions using Markit EDM, Informatica, and AppWorx — standardizing cross-process operations and reducing manual interventions by ~75%.
- Provided production support for daily and monthly data loads and extractions; proactively resolved failures to guarantee uninterrupted feeds into the Data Warehouse.
- Executed unit, integration, and regression test plans; documented technical specifications; collaborated with on-site and offshore teams in Agile via JIRA.
- Key projects: CML Process Owner & PAM JSON Message Generator (Midland → PAM by State Street); Derivatives migration & PAM Legacy Message Generator (Findur → PAM), leading integrity testing, mapping validation, and performance tuning.

Student Assistant | Office of Residence Life IIT – Chicago, IL **August 2021 – May 2023**

- Provided frontline customer service, access-control support, and incident coordination for university residence halls; partnered with resident advisors and professional staff to resolve student concerns and maintain a safe, responsive environment.

Software Engineer Intern | ITHENA – Richmond, VA **June 2020 – August 2020**

- Developed full-stack and IoT components for a connected street-lighting prototype under a senior software engineer in Product Development.
- Built C# application components with MQTT messaging and configured TCP/IP device-to-server communication independent of third-party automation platforms.
- Designed a browser-based monitoring and control interface using JavaScript, HTML, and CSS; authored technical guidelines, test reports, and user documentation.

Education

University of the Cumberlands – Williamsburg, KY (Online) **August 2025 – Present** Ph.D., *Information Technology* | GPA: 3.75 **Expected: May 2029**

Focus: Applied AI, data-driven systems, enterprise automation, and secure/cloud architectures.

Illinois Institute of Technology – Chicago, IL **August 2019 – August 2023** M.S., *Cybersecurity Engineering* | GPA: 3.6 **Awarded: August 2023** B.S., *Computer & Cybersecurity Engineering* | GPA: 3.5

Completed via an accelerated co-terminal program (M.S. begun May 2022 during the final undergraduate year); both degrees conferred August 2023.

Relevant coursework: Application Development, Database Management, Data Structures & Algorithms, Cybersecurity, AI, Machine Learning, Computer Architecture, Embedded Systems, IoT, Network Security, Data Encryption.

Certifications

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| • Investment Foundations Certificate (CFA Institute) | July 2026 |
| • Meta Full-Stack Engineer (Coursera/Meta) | January 2026 |
| • Meta Back-End Developer (Coursera/Meta) | January 2026 |
| • Certificate of Competence in Zero Trust (Cloud Security Alliance) | February 2025 |
| • Meta Front-End Developer (Coursera/Meta) | June 2023 |

- Palo Alto Networks Cybersecurity (**Coursera/Palo Alto**)
- AWS Certified – 4 Certifications Training (**Udemy/BackSpace Academy**)

March 2023

June 2021

Projects

PAM CML Migration & JSON Message Generation Platform | Python, SQL, Markit EDM, PAM Accounting, AppWorx, JSON — *Lead Engineer & IT Owner*

- Designed and built a rules-driven JSON message-generation engine (payments, fees, payoffs, adjustments) supporting 1,500+ active commercial mortgage loans, running daily in production across DEV/UAT/PROD with controlled versioned releases (V1–V7).
- Replaced vendor-managed processing with an internally owned, automated accounting pipeline with built-in source-to-destination reconciliation on every run, eliminating external dependency and improving auditability.

Convo Care — Privacy-First AI Risk Triage Platform | Python, FastAPI, React, MySQL, NLP, Speech-to-Text, Docker — *System Architect & Lead Engineer*

- Built a full-stack platform that analyzes text and voice conversations to produce sentiment and mental-health risk scores, transcribing audio in memory and persisting only numerical scores — zero transcript or raw-audio storage.
- Delivered per-user risk trends, history, and high-risk flagging in a React dashboard, demonstrating privacy-as-architecture for compliant AI decision-support.

Enterprise Event Simulation Platform | Java, Gradle, Distributed Systems, CI — *Designer & Lead Engineer*

- Built a configuration-driven simulator that exercises distributed systems under burst traffic, out-of-order delivery, retries, backpressure, and partial outages before production release.
- Implemented deterministic replay for debugging and regression validation, providing a repeatable testbed for data-correctness and resilience testing.

MQTT Smart Home Controller (Arduino + IoT) | Arduino, C++, MQTT, Embedded Systems — *Designer & Builder*

- Designed modular Arduino firmware with a bidirectional MQTT control plane (publish telemetry, subscribe to commands), integrating DHT sensors and relay modules.
- Demonstrated event-driven, message-based architecture with control/data-plane separation — the same principles applied in enterprise data pipelines.

Plant Disease AI Classifier | Python, Computer Vision, CNN, Edge Computing — *Developer*

- Developed a real-time classifier combining thermal imaging and humidity sensing with a convolutional neural network to detect plant diseases for sustainable agriculture.
- Incorporated Vapor Pressure Deficit (VPD) analytics for early risk detection and actionable irrigation insights.

Publications

Engineering-to-Research Monograph Series — **Zenodo (OpenAIRE-indexed), 2026. Sole author (Palayil, A. B.).**

- Vol. 10 (Capstone) — From Embedded Systems to Explainable AI Governance: A Research Synthesis across Ten Engineering Disciplines. DOI: 10.5281/zenodo.20832713. The capstone consolidates and synthesizes the complete ten-volume series, citing each preceding volume (Vols. 1–9) in full; all ten monographs are openly available in the same Zenodo collection.

Leadership Experience

Internal Secretary | IEEE-Eta Kappa Nu, IIT Delta Chapter – Chicago, IL

April 2022 – May 2023

- Managed agendas and action-item tracking and coordinated professional events, networking mixers, and guest lectures, driving a ~30% increase in member engagement.

Honors & Affiliations: Member, IEEE (2024 – Present); Dean’s List recipient (5 semesters, 2020–2023).